

Programme Health Intelligence

A Learning Analytics Case Study | Megha Sinha | June 2026 | Dataset: OULAD (Open University) | Tools: Python · SQL · Power BI

End-to-end learning analytics project analysing **32,593 student enrolment records** across 7 modules. The analysis identifies the key factors that influence learner success — engagement, assessment performance, and submission behaviour — and translates data into a validated at-risk framework for operations teams.

HEADLINE METRICS

33K Total Enrolments	47.2% Pass Rate	31.2% Withdrawal Rate	21.6% Fail Rate	9.3% Distinction Rate
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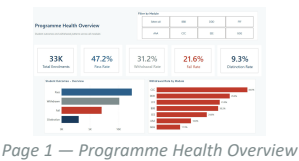
KEY FINDINGS

- 52.8% of enrolments do not result in a successful outcome.
 - Withdrawal rates range from 11.5% (GGG) to 44.5% (CCC) — module-specific, not just student-level.
 - 62.8% of withdrawals occur before the module halfway point; median withdrawal is day 66.
 - 50.2% of withdrawn students never submitted a single assessment — a commitment failure, not performance.
- Students below 75 VLE clicks have under 50% success rate; below 25 clicks, 48.3% withdraw.
 - Assessment score threshold of 55 marks is predictive from week 1 — a 33-point gap in success rates.
 - Late submission predicts poor outcomes independently of score — a standalone behavioural signal.
 - Critical Risk students: 98.7% fail/withdrawal rate vs 16.4% for Low Risk — an 82 percentage point gap.

AT-RISK FRAMEWORK VALIDATION

Risk Category	Students	Success Rate	Fail/Withdrawn Rate
Low Risk	8,825	83.6%	16.4%
Medium Risk	9,312	59.2%	40.8%
High Risk	6,351	37.7%	62.3%
Critical Risk	8,105	1.3%	98.7%

DASHBOARD PREVIEW



TOP OPERATIONAL RECOMMENDATIONS

- Flag students with zero VLE activity by end of week 2 — 7,400 students show no engagement at all.
- Act on first assessment scores below 55 immediately — the signal is reliable from week 1.
- Launch pre-module engagement campaigns — 26.3% of withdrawals happen before the module even starts.
- Prioritise CCC and BBB for intervention — highest combined withdrawal and pre-assessment dropout rates.
- Use the three-signal at-risk framework (engagement + score + submission timing) for weekly monitoring.

Full analysis — 6 Jupyter notebooks, 4 SQL queries, findings report, and interactive Power BI dashboard — available on GitHub. Built with Python (pandas, matplotlib, seaborn), SQLite, and Power BI Desktop.